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Advances in the chemistry of organozinc reagents

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Abstract: In recent years, organozinc compounds have emerged as valuable reagents for transition metal catalyzed cross-couplings and nucleophilic addition reactions. Tolerance of organozincs to common electrophilic functional groups such as ketones, esters and nitriles in combination with intrinsically high reactivity of carbon-zinc bond makes them ideal for the synthesis complex functionalized products. This overview highlights advances in preparation and application of organozinc reagents reported over a period of last six years.

Keywords: Organozinc reagents, Negishi cross-coupling, Transmetalation, Halogen/zinc exchange

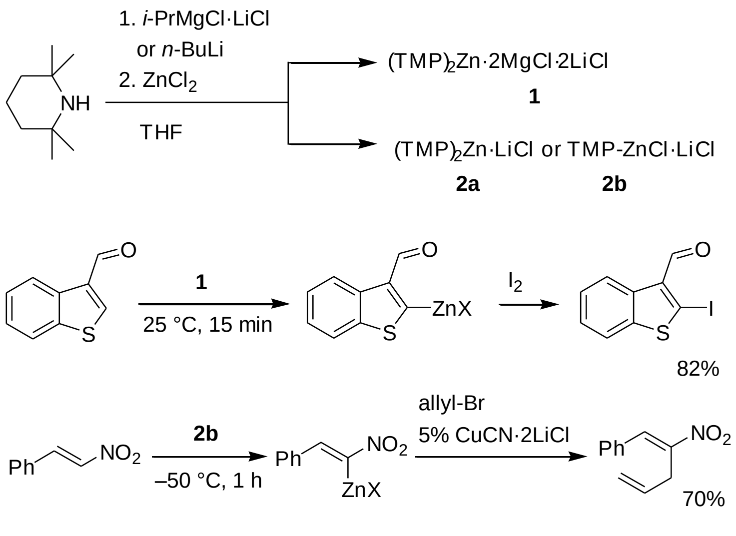
Contents

Introduction

Organozinc reagents have found widespread use in a variety of processes.[1] Compared to Grignard reagents, organozincs offer a unique functional group tolerance coupled with good reactivity. Indeed, ester, keto, cyano, and nitro functions stay unaffected that obviates the use of protective groups. At the same time, carbon-zinc bond readily undergoes transmetalation with transition metals, thereby making organozincs valuable components in cross-coupling reactions. While conventional dialkylzinc reagents are highly pyrophoric, which limits their applications, alkyl (or aryl) zinc halides can be easily handled and stored using conventional Schlenck techniques. Several reviews have been reported within last decade covering earlier achievements.[2] However, the field is developing rapidly, and in the present paper, we cover advances of applications of organozinc reagents over a period of six years.

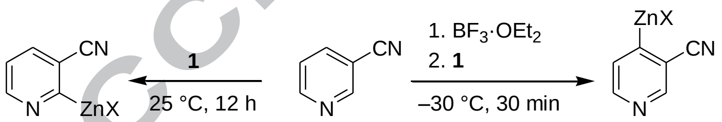
Preparation

Preparation of organozinc reagents by treatment of the C-H function with a zinc base has long been underexplored, apparently, due to moderate kinetic basicity of element-zinc bonds. The successful application of this concept was realized, in a general way, by using zinc amides of 2,2,6,6-tetramethylpiperidine (TMP-H) **1** and **2a,b** generated from TMP-magnesium chloride[3a] or TMP-lithium[3b] and zinc chloride (Scheme 1). An alternative method to prepare TMPZnCl·LiCl is to effect insertion of zinc into N-Cl bond of TMPCl.[4] Using reagent **1**, zincation of various aromatic and heteroaromatic compounds was effected.[3a] This methodology allows for the generation of organozincs bearing unprotected aldehyde group, which also remains unaffected upon subsequent reactions with electrophiles. Reagents **2a,b** were employed for metalation of electron-deficient alkenes. For example, β-nitrostyrene, which itself is a reactive Michael acceptor, was metalated with zinc amide **2b** followed by copper-catalyzed allylation.[3b]



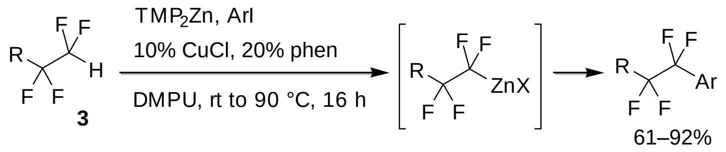
Scheme 1. Metalation of sp^2 C-H bonds.

Direct zincation of unactivated pyridines is inefficient. However, the presence of electron withdrawing cyano group allows for a slow zincation at room temperature (Scheme 2). The reactivity of pyridines towards metalation can be dramatically increased by complexation with boron trifluoride, as demonstrated for 3-cyanopyridine.[5] Besides rate acceleration, complexation incurs switch of regioselectivity. The preferred abstraction of hydrogen from position 4 of complexed 3-cyanopyridine may be associated with steric reasons.



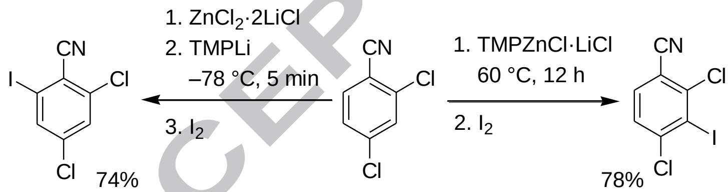
Scheme 2. Zincation of 3-cyano pyridine.

Coupling of fluorinated alkanes **3** with aromatic iodides was performed using TMP_2Zn in 1,3-dimethylpropyleneurea (DMPU) as solvent[6] (Scheme 3). The increased C-H acidity of these substrates is defined by electronwithdrawing fluorine atoms. For these reactions, zinc base TMP_2Zn, which is free from lithium chloride, has to be employed. Zinc base serves to abstract the proton to generate organozinc species with subsequent zinc/copper transmetalation and arylation reaction. Despite tedious procedure for removal of LiCl from zinc amide, overall protocol constitutes an attractive way for direct one-step conversion of readily available fluorinated alkanes into various functionalized products.



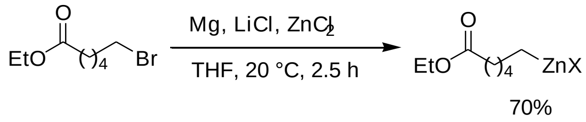
Scheme 3. Metalation of sp^3 C-H bonds.

Lithium bases are notably more reactive than zinc analogues, but organolithiums can be notably less stable compared to organozinc counterparts. Correspondingly, rapid generation of organolithium spices using lithium amide (LiTMP) can be coupled with its rapid trapping by the present zinc dichloride to effect lithium/zinc exchange[7] (Scheme 4, left arrow). Of special note is that the regioselectivity of metalation using lithium amide is different to that of using zinc amide (Scheme 4, right arrow).



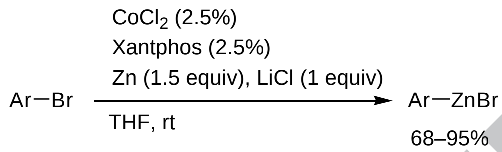
Scheme 4. Metalation of 2,4-dichlorobenzonitrile.

Reaction of organic halides with elemental zinc is a typical method for making organozincs. However, for some substrates the reaction is slow, and harsh conditions or application of reactive form of zinc (Rieke-zinc) may be required. Though organozincs may be prepared from Grignard reagents by magnesium/zinc exchange, functionalized organomagnesiums are difficult to prepare owing to their limited stability. In this case, a procedure relying on insertion of magnesium into carbon-halogen bond with immediate magnesium/zinc exchange was developed.[8] For example, the reaction of an alkyl bromide bearing ester group using magnesium in the presence of zinc chloride proceeded rapidly at room temperature affording the organozinc species (Scheme 5).



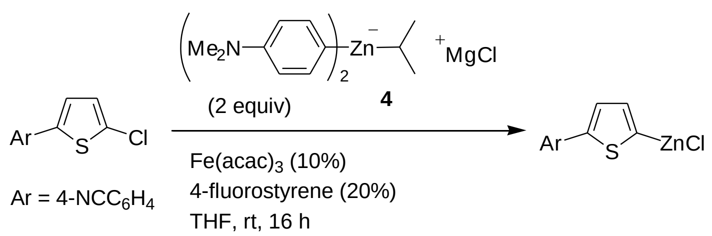
Scheme 5. Preparation of ester-substituted organozinc reagent.

Alternative opportunity to accelerate reaction with zinc is to use a transition metal which would react with organic halide by means of oxidative addition followed by metal/zinc transmetalation. Early work involved ligandless cobalt(II) bromide/zinc system, which is particularly suitable for aromatic bromides.[9] Recently, another cobalt-catalyzed process was developed based on cobalt(II) chloride in combination with Xantphos ligand[10] (Scheme 6).



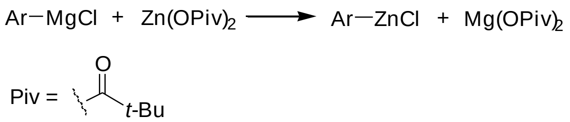
Scheme 6. Co-catalyzed reduction.

Halogen/zinc exchange is a popular approach to organozincs. However, the method is most efficient for aryl iodides and for certain bromides, whereas aryl chlorides are typically unreactive. To effect chlorine/zinc exchange a couple of zincate reagents under the catalysis of iron or cobalt salts were proposed[11] (Scheme 7). For example, thienyl chloride was converted into corresponding zinc chloride using two equivalents of zincate **4** along with catalytic amounts of iron(III) acetylacetonate and 4-fluorostyrene. Importantly, this procedure was also applied for the generation of organozincs from alkyl chlorides.



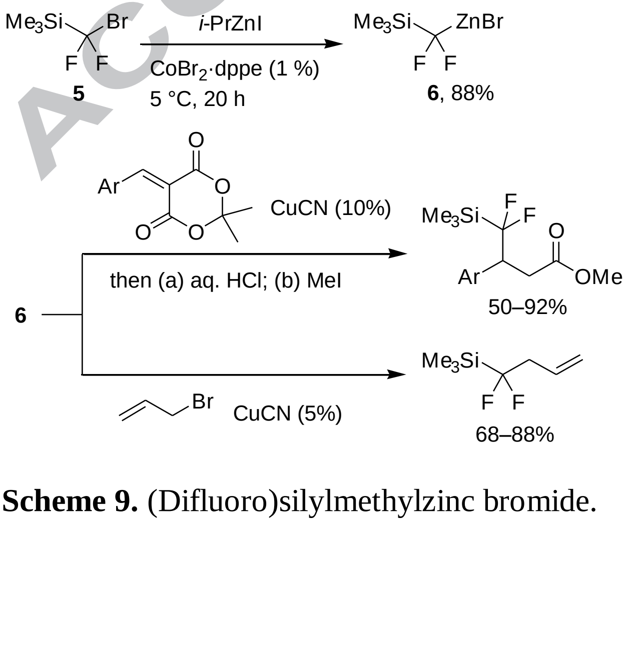
Scheme 7. Chlorine/zinc exchange.

Organozinc reagents are well known for their air-sensitivity. To address this very important issue, a reagents obtained by reaction of organomagnesiums with zinc pivalate were evaluated[12] (Scheme 8). Studies on the composition of this system suggested the separate existence of aryl zinc chloride and magnesium pivalate. Nevertheless, these modified reagents are notably less sensitive towards air. The latter phenomenon was explained by ability of magnesium pivalate to protect organozinc reagents by sequestering water and oxygen contaminants.



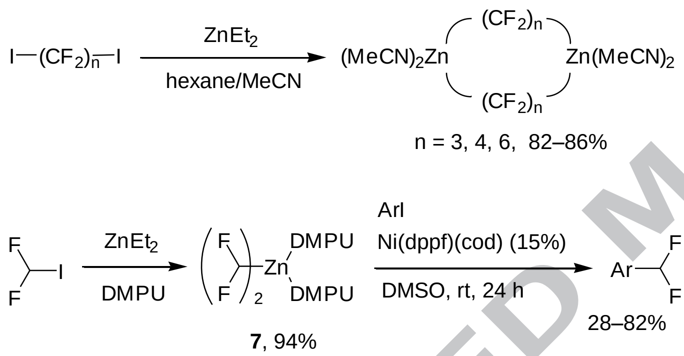
Scheme 8. Reagents modified by magnesium pivalate.

Bromine/zinc exchange in silane **5** was effected using isopropylzinc iodide catalyzed by 1 mol % of cobalt(II) bromide-dppe complex[13] (Scheme 9). It was established that reagent **6** at zinc contains bromine, but not iodine, which was explained by a mechanism involving cobalt(I) iodide as a catalytically active species. In the presence of copper cyanide reagent **6** was cross-coupled with allyl bromides[13] and added at C=C bond of arylidene derivatives of Meldrum's acid.[14]



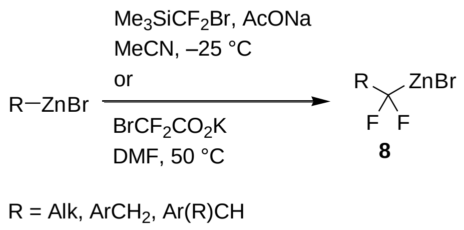
Scheme 9. (Difluoro)silylmethylzinc bromide.

Fluorinated organozincs can be accessed by halogen/zinc exchange using diethylzinc.[15] Application of this method to perfluorinated diiodides leads to dimeric organozincs, which were employed in copper-catalyzed couplings with 1,2-diiodoaromatics affording fused fluorinated products[16] (Scheme 10). A similar metalation reaction was used to prepare bis(difluoromethyl)zinc complexed with DMPU. Reagent **7** is a stable solid storable for months under inert atmosphere. Its coupling with aromatic iodides is catalyzed by a nickel complex.[17] It should be pointed out that this coupling proceeds at room temperature, whereas similar processes involving intermediate generation of difluoromethyl copper require heating above 100 °C.



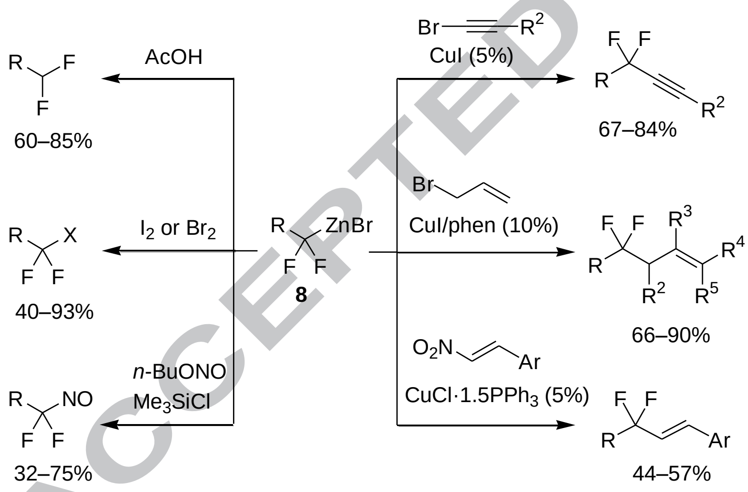
Scheme 10. Iodine/zinc exchange.

Recently, a general approach for the preparation of *gem*-difluorinated organozincs was proposed based on insertion of CF_2-fragment into carbon-zinc bond[18] (Scheme 11). As a source of difluormethylene unit, (bromodifluoromethyl)trimethylsilane[18a] or potassium bromodifluoroacetate[18b] were employed. It is likely that reaction involves generation of difluorocarbene which undergoes insertion into C-Zn bond, but the mechanism of insertion is not clear at present. The reactions works well for primary and secondary alkyl and benzyl organozincs. Reagents **8** possess limited stability, and, when kept in acetonitrile are prone to decomposition within few hours at room temperature. However, their stability can be markedly increased by addition of at least two equivalents of dimethylformamide.



Scheme 11. Insertion of CF_2-fragment.

Reagents **8** react with various electrophiles (Scheme 12). Their protonation affords products with the difluoromethyl group, while reaction with iodine or bromine leads to compounds with CF_2I and CF_2Br substituents. Nitrosation was performed using butyl nitrite/chlorosilane system affording *gem*-difluoronitroso compounds.[18d] Reagents **8** were also coupled with allyl bromides[18c] and 1-bromoalkynes[18f] in the presence of copper(I) catalysts. Upon reaction with nitrostyrenes, instead of expected conjugate addition, substitution of the nitro group by a fluorinated fragment occurred.[18g] The latter process likely proceeds by a free radical mechanism.

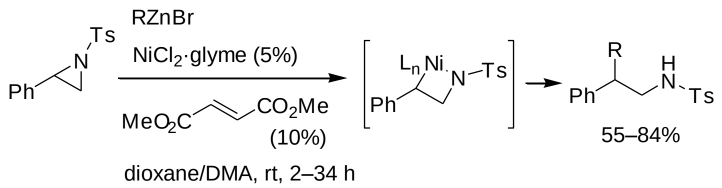


Scheme 12. Chemistry of reagents **8** (yields are based on starting RZnBr).

Cross-coupling reactions

Organozincs are excellent reagents for transition-metal mediated processes, and Negishi cross-coupling enjoyed numerous applications. Not surprisingly, contributions from many laboratories aimed at the extending the scope of the reaction have been reported in recent times.

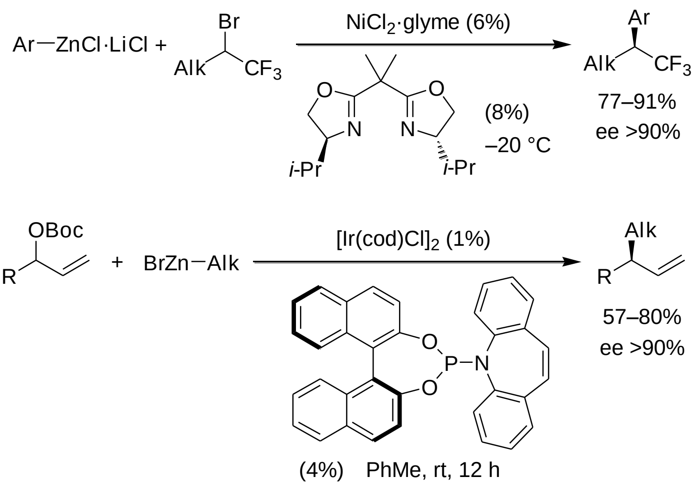
N-Tosylaziridines were used as electrophilic components in coupling with alkylzinc bromides[19] (Scheme 13). The first step is oxidative addition of low-valent nickel species into C-N bond of the aziridine. This step may proceed either via S_N2 type substitution or via radical pathway. At the final step, transmetalation and reductive elimination completes the catalytic cycle. The presence of dimethylfumarate was essential for the reaction to proceed. It was proposed that this electron deficient alkene serves as a ligand to nickel and accelerates reductive elimination.



Scheme 13. Reactions of aziridines.

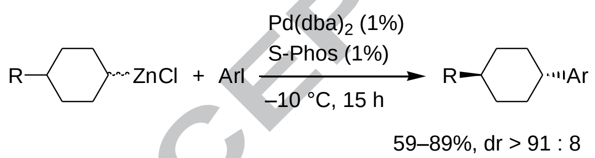
Arylation of α-CF_3-substituted alkyl bromides was performed using aromatic organozinc regents obtained from aryllithiums and zinc chloride[20] (Scheme 14). The reaction is catalyzed by nickel dichloride along with chiral bisoxazoline ligand. Notably, racemic secondary bromides stereoconvergently afforded arylated products with excellent enantioselectivities. The stereoconvergency is associated either with intermediate formation of a radical during oxidative addition step or with interconversion of alkylnickel species.

Asymmetric allylation of alkyl organozincs with racemic Boc-protected secondary allylic alcohols was also performed in a stereoconvergent fashion.[21] Iridium catalyst in combination with monodentate BINIOL-derived phosphorus chiral ligand was employed.



Scheme 14. Stereodivergent reactions.

Cyclohexylzinc chlorides couple with aromatic iodides in the presence of palladium catalyst and a Buchwald-type ligand S-Phos[22] (Scheme 15). Strating zinc reagents were prepared from Grignard regents and were employed as mixtures of diastereoisomers. However, coupling products were produced predominantly with *trans* (for *para* and *ortho* isomers) or *cis* (for *meta* isomer) arrangement. The observed stereochemical outcome was explained by equilibration of organozinc reagent and the preferential formation of the most stable isomer of palladium(II) intermediate generated after transmetalation step.

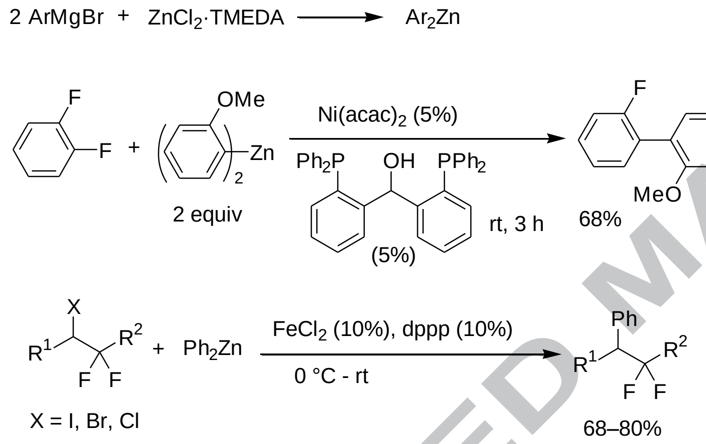


Scheme 15. Arylation of cyclohexylzinc chlorides.

Diarylzinc reagents can be generated *in situ* from Grignard reagents and zinc chloride. However, highly hygroscopic nature of zinc chloride makes its handling inconvenient. For this purpose, zinc chloride coordinated with *N*,*N*,*N'*,*N'*-tetramethylethylenediamine (TMEDA) or diglyme can be used.

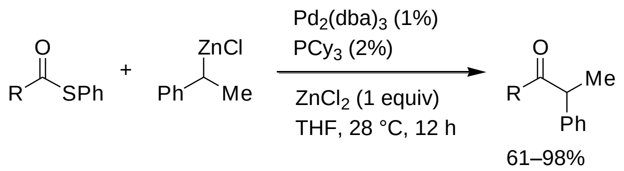
Aryl fluorides are difficult partners for cross-couplings because of strong carbon-fluorine bond. For polyfluorinated arenes the problem is further complicated by the possibility for substitution of several fluorides. A diphosphine ligand bearing a hydroxy group, which under reaction conditions is transformed into alkoxide, was effective in combination with a nickel catalyst[23] (Scheme 16). The binding of the alcoxide to nickel was proposed to be important for reaction efficiency. Though product yields were in the order of 60-70%, the selectivity for monosubstitution in polyfluorinated substrates was quite good, and even monosubstitution of hexafluorobenzene was achieved.

Reaction of α-*gem*-difluorinated alkyl halides with diarylzinc is mediated by catalytic amounts of iron(II) chloride and bis(diphenylphosphino)propane.[24] Mechanistic studies demosntrated that the reaction proceeds by a free radical pathway.



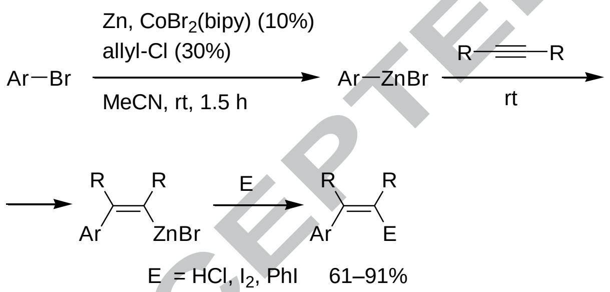
Scheme 16. Reactions of Ar_2Zn generated from $ZnCl_2$·TMEDA.

Acylating reagents serve as good electrophiles for palladium catalyzed cross-coupling. However, the use of secondary alkylzinc reagents in this process was limited. Reaction of thioesters with organozincs, known as Fukuyama cross-coupling, is characterized by very mild conditions and high functional group tolerance, and, therefore, can be utilized for constructing complex functionalized molecules. Reactions of phenylthioesters with secondary benzyl zinc reagents was catalyzed by palladium complex and tri(cyclohexyl)phosphine[25] (Scheme 17). It should be pointed out that the addition of one equivalent of zinc chloride provided significant rate acceleration and allowed decrease of amount of palladium. The effect of zinc chloride may be to shift the Schlenk equilibrium or to activate palladium catalyst.



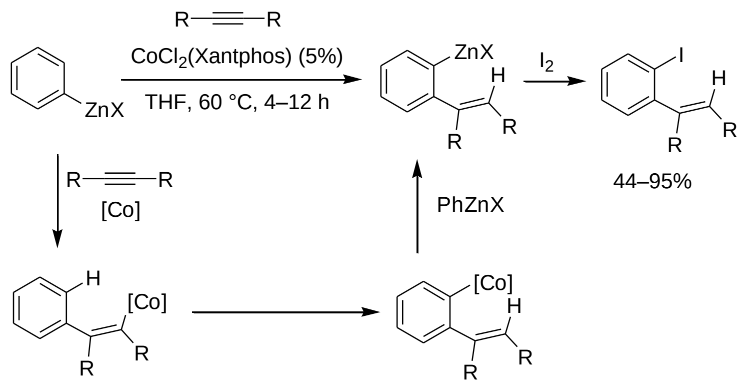
Scheme 17. Fukuyama coupling.

While reaction of aromatic bromides and iodides with elemental zinc may require polar coordinating solvents or elevated temperatures,[26] the rate of these processes may be increased by addition of transition metal catalysts[9,10] (see also Scheme 6). Besides accelerating zinc insertion, these catalysts may promote further transformations. Thus, reaction of aryl bromides with zinc catalyzed by cobalt(II) bromide-bi(pyridyl) complex in acetonitrile afforded arylzinc reagents[27] (Scheme 18). At this stage, the mixture must be filtered to remove excess of unreacted zinc, and the arylzinc bromide undergoes carbozincation of internal alkynes. The resulting vinylzinc bromide may be quenched by acid or iodine or cross-coupled with phenyl iodide.



Scheme 18. Cobalt-catalyzed processes.

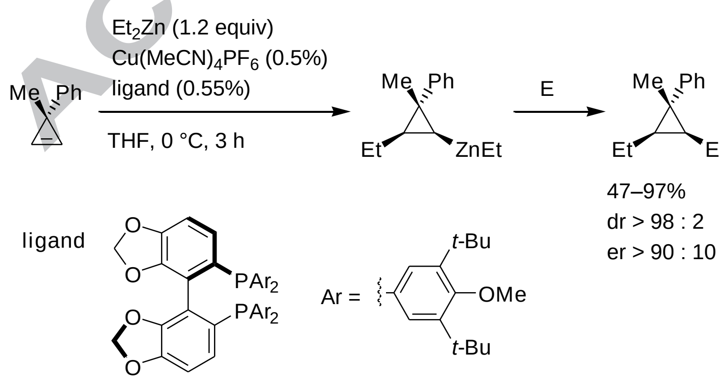
When reaction of arylzinc halides with alkynes is performed in the presence of cobalt(II) chloride-Xantphos complex in warm tetrahydrofuran, the carbozincation of triple bond was followed by *ortho*-C-H-activation[28] (Scheme 19). The reaction presumably proceeds by exchange of proximal C-H and C-Co bonds with subsequent cobalt/zinc transmetalation. The *ortho*-fucntionalized arylzinc species may be iodinatied or subjected to other reactions typical for organozincs. Similar migration of cobalt was observed for coupling of arylzinc halides with norbornene.[29]



Scheme 19. *Ortho*-C-H activation

Reactions of alkyl halides with terminal alkynes and zinc mediated by iron(II) salts leading to vinyl zinc species were reported.[30] While these reactions can be formally described as *anti*-carbozincation of the triple bond, it was proposed that they proceed as iron triggered radical addition, and, therefore, alkyl organozincs are not involved.

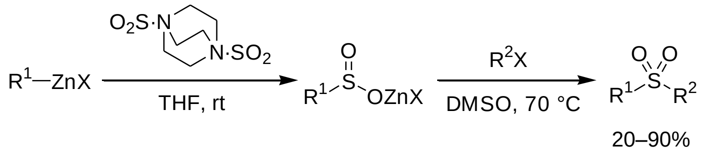
Dialkylzincs add at double bond of cyclopropenes in a *syn*-mode in the presence of a copper(I) catalyst and a chiral phosphine ligand[31] (Scheme 20). The resulting cyclopropyl zinc reagents couple effectively with various electrophiles with retention of configuration affording products with excellent diastereoselectivities and good to high enantioselectivities.



Scheme 20. Adition to cyclopropenes.

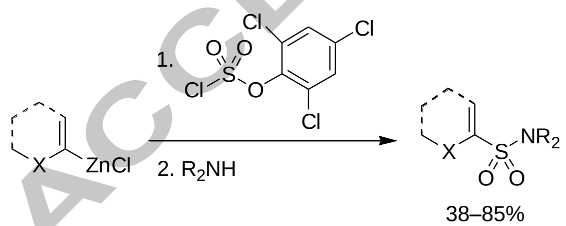
Reactions with heteroatom electrophiles

Sulfur dioxide is a good electrophile for coupling with polar organometallics. At the same time, it is a toxic gas, which renders its applications inconvenient. Sulfur dioxide forms a crystalline adduct with diazabicyclooctane with the stoichiometry of 2 moles of SO_2 per mole of diazabicyclooctane. This reagent serves as an equivalent of SO_2 in reactions with organozincs[32] (Scheme 21). The reaction proceeds at room temperature affording sulfinate salt which can be further converted to sulfones by adding dimethylsulfoxide co-solvent and an alkylating agent.



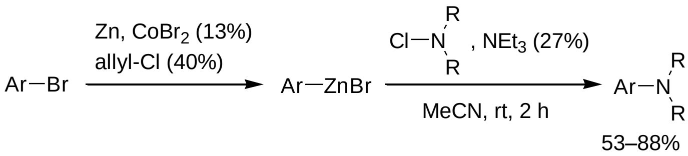
Scheme 21. Synthesis of sulfones.

Hetaryl zinc reagents may be coupled with 2,4,6-trichlorophenyl chlorosulfate with subsequent reaction with amines furnishing sulfonamides[33] (Scheme 22). After first step, products of substitution of either chlorine of trichlorophenoxy group are produced depending on the type of organozinc reagent. It was proposed that even in the former case the substitution of trichlorophenoxy group takes place followed by reaction of sulfochloride with zinc phenoxide.



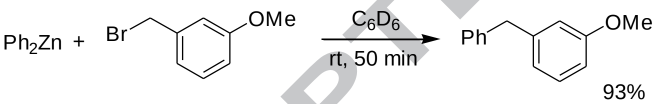
Scheme 22. Synthesis of sulfonamides.

Arylzinc bromides, obtained from aryl bromides and zinc with cobalt(II) bromide as catalyst, react with *N*-chloroamines leading to arylamines[34] (Scheme 23). Substoichiometric amount of triethylamine has to be added to suppress side reactions such as chlorination and protonation of organozinc reagent. Reaction has a wide scope — cyclic and acyclic *N*-chloroamines give good yields of products. It was suggested that the reaction mechanism involves cobalt(I) species with its oxidative addition into N-Cl bond, transmetalation and reductive elimination.



Scheme 23. Syntheis of arylamines.

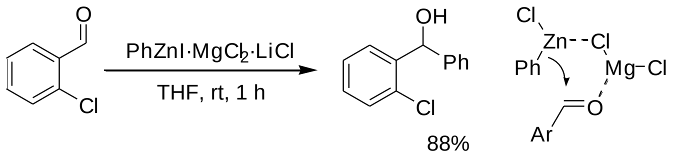
Diaryl zinc reagents couple with benzyl bromides and tertiary and secondary alkyl bromides in the absence of transition metal catalysts within few minutes at room temperature[35] (Scheme 24). Primary alkyl bromides require heating at 80 °C for 20 hours. The mixture should not contain coordinating ethereal solvents, since the latter drastically inhibit the reaction. The rate of coupling increases with Lewis acidity of diaryl zinc. At the same time, mechanistic studies suggested the intermediacy of radical species.



Scheme 24. Uncatalyzed coupling.

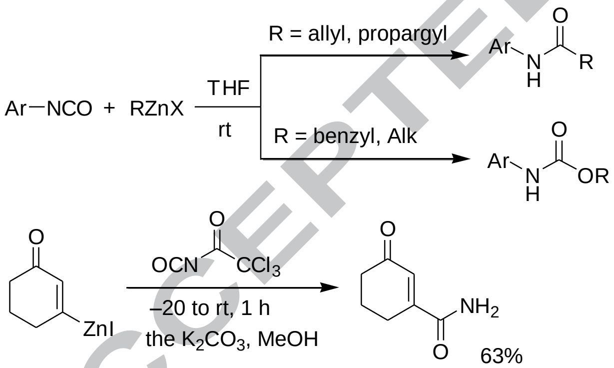
Addition to double bonds

Organozinc chlorides are typically poorly reactive towards carbonyl compounds. For example, phenylzinc iodide reacted with *ortho*-chlorobenazldehyde within 72 h leading to addition product with only 60% yield. Similar reaction performed with the organozinc containing one equivalent of $MgCl_2$ (organozinc was prepared by the reaction of phenyl iodide with magnesium/zinc/lithium chloride system) was complete within 1 h and afforded the product in 88% yield[36] (Scheme 25). It was proposed that magnesium chloride takes part in six-membered transition state for the addition event and plays a role of Lewis acid for activation of the carbonyl group.



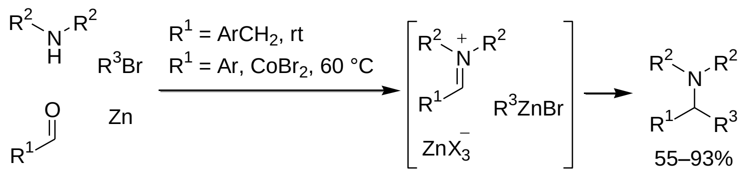
Scheme 25. Addition to aldehydes.

Reaction of aryl isocyantes with organozinc reagents provides different products depending on the structure of organozinc[37] (Scheme 26). Thus, allyl and propargyl zinc bromides added at the C=O bond leading to amides as anticipated. On the contrary, reactions of benzyl and alkyl zincs with the same isocyanates under similar conditions afforded carbamates with the yields around 60%. The mechanism of the latter reaction, as well as the source of the oxygen atom, is unclear. Trichloroacetyl isocyanate is particularly electrophilic owing to electron accepting trichloroacetyl moiety. It rapidly reacts with aryl and vinyl organozincs, and subsequent methanolysis affords primary amides.[38]



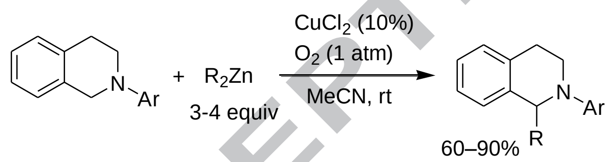
Scheme 26. Reactions of isocyanates.

Compared to carbonyl compounds, iminium ions are more reactive electrophiles. Though iminium salts can be frequently prepared in isolated form, they are typically generated *in situ*. Their equilibrium formation from aldehydes and secondary amines opens opportunities for three component couplings using a suitable nucleophile. Organozincs are good candidates for such couplings, since they are poorly reactive towards aldehydes, but sufficiently nucleophilic to interact with iminium species. Furthermore, organozincs can also be prepared directly from organic bromides and zinc[39] (Scheme 27). In acetonitrile, benzyl bromides are reduced with zinc rapidly at room temperature, while for aromatic bromides, the reaction require addition of cobalt bromide and heating. Another issue is that excess amount of organic bromide has to be used (at least 2.5 equiv), likely to compensate for acidic hydrogen arising upon formation of iminium ion.



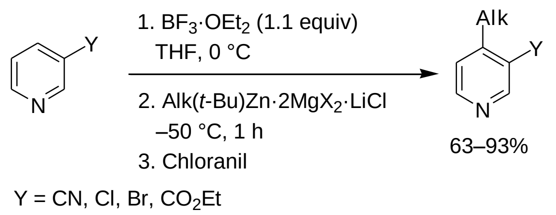
Scheme 27. Three-component coupling.

Iminium ions can be generated by other means. For examples, C-H oxidation of tetrahydroisoquinolines with copper(II) and oxygen provides iminium species which is trapped by dialkylzinc[40] (Scheme 28). The organometallic reagent was used in excess amount.



Scheme 28. Reactions of tetrahydroisoquinolines.

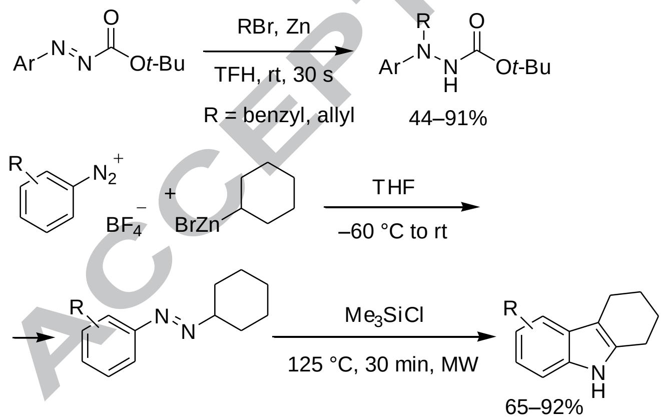
Complexation of pyridine with Lewis acid significantly increases electrophilicity of the aromatic ring. *Meta*-substituted pyridines compelxed with boron trifluoride react with diorganozincs at –50 °C (Scheme 29). The primary addition product was oxidized with chloranil (tetrachloroquinone) affording 4-substituted pyridine.[41] In this reaction, mixed organozinc reagent, bearing *tert*-butyl as a non-transferable group, was used which was prepared from alkylzinc bromide (obtained using Mg/$ZnCl_2$/LiCl system) and *tert*-butyl magnesium chloride.



Scheme 29. Addition to pyridines.

Benzyl and allyl bromides react with azocarboxylic esters and zinc under Barbier conditions. Reaction was complete within one minute at room temperature, likely because of facile formation of benzyl and allyl organozinc reagents[42] (Scheme 30). The nucleophilic addition proceeded exclusively at nitrogen bearing aryl group, which is associated with effective stabilization of incipient negative charge by the carbonyl group.

Diazonium salts readily couple with organozincs. If alkyl organozinc reagents are used, the resulting azo compound can be involved into Fischer indole synthesis by microwave heating in the presence of chlorosilane.[43] Use of cyclic organozincs, such as cyclohexyl zinc bromide, led to fused indole compounds.



Scheme 30. Addition to azo compounds.

Conclusions

Over the last six years significant advances in organozinc chemistry have been achieved. The scope of available organozinc reagents and methods for their generation have been expanded. Novel approaches based on direct zincation of C-H bonds, as well as *in situ* magnesium/zinc transmetalation, allowed straightforward access to those reagents, which for a long time have been considered problematic. Concerning application of organozincs, the cross-coupling protocols are improving gradually. Of special note is the advent of stereoconvergent couplings which eliminates the need to take care of configurational stability of organometallics. The chemistry of fluorinated organozincs has also suffered significant progress, which is a reflection of importance of organofluorine compounds.

Acknowledgments

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References

(1) (a) *The Chemistry of Organozinc Compounds*; Rappoport, Z., Marek, I., Eds.; John Wiley & Sons: Chichester, U.K., 2006. (b) Knochel, P.; Leuser, H.; Gong, L. Z.; Perrone, S.; Kneisel, F. F. In *Handbook of Functionalized Organometallics*; Wiley-VCH: Weinheim, 2005, p 251–346. (c) Knochel, P.; Calaza, M. I.; Hupe, E.; de Meijere, A.; Diederich, F. In *Metal-Catalyzed Cross-Coupling Reactions*; Wiley-VCH: Weinheim, 2004, p 619–670. (c) Knochel, P.; Millot, N.; Rodrigues, A. L. *Org. React. (N.Y.)* **2001**, *58*. (d) Knochel, P.; Singer, R. D. *Chem. Rev.* **1993**, *93*, 2117–2188.

(2) (a) Knochel, P. In *Organometallics in Synthesis*; John Wiley & Sons, Inc.: 2013, p 223–372. (b) Chemla, F.; Ferreira, F.; Jackowski, O.; Micouin, L.; Perez-Luna, A. In *Metal-Catalyzed Cross-Coupling Reactions and More*; Wiley-VCH Verlag GmbH & Co. KGaA: 2014, p 279–364. (c) Dagousset, G.; Francois, C.; Leon, T.; Blanc, R.; Sansiaume-Dagousset, E.; Knochel, P. *Synthesis* **2014**, *46*, 3133–3171. (d) Kim, J. H.; Ko, Y. O.; Bouffard, J.; Lee, S. *Chem. Soc. Rev.* **2015**, *44*, 2489–2507. (e) Sidduri, A.; Tilley, J. W.; Fotouhi, N. *Synthesis* **2014**, *46*, 430–444. (f) Murakami, K.; Yorimitsu, H. *Beilstein J. Org. Chem.* **2013**, *9*, 278–302.

(3) (a) Wunderlich, S. H.; Knochel, P. *Angew. Chem. Int. Ed.* **2007**, *46*, 7685–7688. (b) Bresser, T.; Knochel, P. *Angew. Chem. Int. Ed.* **2011**, *50*, 1914–1917.

(4) Unsinn, A.; Ford, M. J.; Knochel, P. *Org. Lett.* **2013**, *15*, 1128–1131.

(5) Jaric, M.; Haag, B.; Unsinn, A.; Karaghiosoff, K.; Knochel, P. *Angew. Chem. Int. Ed.* **2010**, *49*, 5451–5455.

(6) Popov, I.; Lindeman, S.; Daugulis, O. *J. Am. Chem. Soc.* **2011**, *133*, 9286–9289.

(7) Frischmuth, A.; Fernández, M.; Barl, N. M.; Achrainer, F.; Zipse, H.; Berionni, G.; Mayr, H.; Karaghiosoff, K.; Knochel, P. *Angew. Chem. Int. Ed.* **2014**, *53*, 7928–7932.

(8) (a) Blumke, T. D.; Piller, F. M.; Knochel, P. *Chem. Commun.* **2010**, *46*, 4082–4084. (b) Piller, F. M.; Metzger, A.; Schade, M. A.; Haag, B. A.; Gavryushin, A.; Knochel, P. *Chem. Eur. J.* **2009**, *15*, 7192–7202. (c) Shen, Z.-L.; Knochel, P. *ACS Catalysis* **2015**, *5*, 2324–2328.

(9) Fillon, H.; Gosmini, C.; Perichon, J. *J. Am. Chem. Soc.* **2003**, *125*, 3867–3870.

(10) Jin, M.-Y.; Yoshikai, N. *J. Org. Chem.* **2011**, *76*, 1972–1978.

(11) Melzig, L.; Diene, C. R.; Rohbogner, C. J.; Knochel, P. *Org. Lett.* **2011**, *13*, 3174–3177.

(12) (a) Bernhardt, S.; Manolikakes, G.; Kunz, T.; Knochel, P. *Angew. Chem. Int. Ed.* **2011**, *50*, 9205–9209. (b) Stathakis, C. I.; Bernhardt, S.; Quint, V.; Knochel, P. *Angew. Chem. Int. Ed.* **2012**, *51*, 9428–9432. (b) Hernán-Gómez, A.; Herd, E.; Hevia, E.; Kennedy, A. R.; Knochel, P.; Koszinowski, K.; Manolikakes, S. M.; Mulvey, R. E.; Schnegelsberg, C. *Angew. Chem. Int. Ed.* **2014**, *53*, 2706–2710.

(13) Kosobokov, M. D.; Levin, V. V.; Zemtsov, A. A.; Struchkova, M. I.; Korlyukov, A. A.; Arkhipov, D. E.; Dilman, A. D. *Org. Lett.* **2014**, *16*, 1438–1441.

(14) Volodin, A. D.; Zemtsov, A. A.; Levin, V. V.; Struchkova, M. I.; Dilman, A. D. *J. Fluorine Chem.* **2015**, *176*, 57–60.

(15) Naumann, D.; Schorn, C.; Tyrra, W. *Z. Anorg. Allg. Chem.* **1999**, *625*, 827–830.

(16) (a) Kaplan, P. T.; Xu, L.; Chen, B.; McGarry, K. R.; Yu, S.; Wang, H.; Vicic, D. A. *Organometallics* **2013**, *32*, 7552–7558. (b) Kaplan, P. T.; Chen, B.; Vicic, D. A. *J. Fluorine Chem.* **2014**, *168*, 158–162.

(17) Xu, L.; Vicic, D. A. *J. Am. Chem. Soc.* **2016**, *138*, 2536–2539.

(18) (a) Levin, V. V.; Zemtsov, A. A.; Struchkova, M. I.; Dilman, A. D. *Org. Lett.* **2013**, *15*, 917–919. (b) Levin, V. V.; Zemtsov, A. A.; Struchkova, M. I.; Dilman, A. D. *J. Fluorine Chem.* **2015**, *171*, 97–101. (c) Zemtsov, A. A.; Kondratyev, N. S.; Levin, V. V.; Struchkova, M. I.; Dilman, A. D. *J. Org. Chem.* **2014**, *79*, 818–822. (d) Smirnov, V. O.; Struchkova, M. I.; Arkhipov, D. E.; Korlyukov, A. A.; Dilman, A. D. *J. Org. Chem.* **2014**, *79*, 11819–11823. (e) Smirnov, V. O.; Maslov, A. S.; Levin, V. V.; Struchkova, M. I.; Dilman, A. D. *Russ. Chem. Bull., Int. Ed.* **2014**, *63*, 2564–2566. (f) Zemtsov, A. A.; Volodin, A. D.; Levin, V. V.; Struchkova, M. I.; Dilman, A. D. *Beilstein J. Org. Chem.* **2015**, *11*, 2145–2149. (g) Kondratyev, N. S.; Levin, V. V.; Zemtsov, A. A.; Struchkova, M. I.; Dilman, A. D. *J. Fluorine Chem.* **2015**, *176*, 89–92. (h) Dilman, A. D.; Levin, V. V. *Mendeleev Commun.* **2015**, *25*, 239–244.

(19) Huang, C.-Y.; Doyle, A. G. *J. Am. Chem. Soc.* **2012**, *134*, 9541–9544.

(20) Liang, Y.; Fu, G. C. *J. Am. Chem. Soc.* **2015**, *137*, 9523–9526.

(21) Hamilton, J. Y.; Sarlah, D.; Carreira, E. M. *Angew. Chem. Int. Ed.* **2015**, *54*, 7644–7647.

(22) Thaler, T.; Haag, B.; Gavryushin, A.; Schober, K.; Hartmann, E.; Gschwind, R. M.; Zipse, H.; Mayer, P.; Knochel, P. *Nat Chem* **2010**, *2*, 125–130.

(23) Nakamura, Y.; Yoshikai, N.; Ilies, L.; Nakamura, E. *Org. Lett.* **2012**, *14*, 3316–3319.

(24) Lin, X.; Zheng, F.; Qing, F.-L. *Organometallics* **2012**, *31*, 1578–1582.

(25) Cherney, A. H.; Reisman, S. E. *Tetrahedron* **2014**, *70*, 3259–3265.

(26) Krasovskiy, A.; Malakhov, V.; Gavryushin, A.; Knochel, P. *Angew. Chem. Int. Ed.* **2006**, *45*, 6040–6044.

(27) Corpet, M.; Gosmini, C. *Chem. Commun.* **2012**, *48*, 11561–11563.

(28) Tan, B.-H.; Dong, J.; Yoshikai, N. *Angew. Chem. Int. Ed.* **2012**, *51*, 9610–9614.

(29) Tan, B.-H.; Yoshikai, N. *Org. Lett.* **2014**, *16*, 3392–3395.

(30) (a) Cheung, C. W.; Zhurkin, F. E.; Hu, X. *J. Am. Chem. Soc.* **2015**, *137*, 4932–4935. (b) Cheung, C. W.; Hu, X. *Chem. Eur. J.* **2015**, *21*, 18439–18444.

(31) Müller, D. S.; Marek, I. *J. Am. Chem. Soc.* **2015**, *137*, 15414–15417.

(32) Rocke, B. N.; Bahnck, K. B.; Herr, M.; Lavergne, S.; Mascitti, V.; Perreault, C.; Polivkova, J.; Shavnya, A. *Org. Lett.* **2014**, *16*, 154–157.

(33) Colombe, J. R.; DeBergh, J. R.; Buchwald, S. L. *Org. Lett.* **2015**, *17*, 3170–3173.

(34) Qian, X.; Yu, Z.; Auffrant, A.; Gosmini, C. *Chem. Eur. J.* **2013**, *19*, 6225–6229.

(35) Dunsford, J. J.; Clark, E. R.; Ingleson, M. J. *Angew. Chem. Int. Ed.* **2015**, *54*, 5688–5692.

(36) Metzger, A.; Bernhardt, S.; Manolikakes, G.; Knochel, P. *Angew. Chem. Int. Ed.* **2010**, *49*, 4665–4668.

(37) Yang, H.; Huang, D.; Wang, K.-H.; Xu, C.; Niu, T.; Hu, Y. *Tetrahedron* **2013**, *69*, 2588–2593.

(38) Schade, M. A.; Manolikakes, G.; Knochel, P. *Org. Lett.* **2010**, *12*, 3648–3650.

(39) (a) Le Gall, E.; Haurena, C.; Sengmany, S. p.; Martens, T.; Troupel, M. *J. Org. Chem.* **2009**, *74*, 7970–7973. (b) Haurena, C.; Le Gall, E.; Sengmany, S.; Martens, T.; Troupel, M. *J. Org. Chem.* **2010**, *75*, 2645–2650. (c) Le Gall, E.; Sengmany, S.; Haurena, C.; Leonel, E.; Martens, T. *J. Organomet. Chem.* **2013**, *736*, 27–35.

(40) Wang, T.; Schrempp, M.; Berndhaeuser, A.; Schiemann, O.; Menche, D. *Org. Lett.* **2015**, *17*, 3982–3985.

(41) Chen, Q.; du Jourdin, X. M.; Knochel, P. *J. Am. Chem. Soc.* **2013**, *135*, 4958–4961.

(42) Lasch, R.; Heinrich, M. R. *J. Org. Chem.* **2015**, *80*, 10412–10420.

(43) Haag, B. A.; Zhang, Z.-G.; Li, J.-S.; Knochel, P. *Angew. Chem. Int. Ed.* **2010**, *49*, 9513–9516.

Graphical abstract

Advances in the chemistry of organozinc reagents

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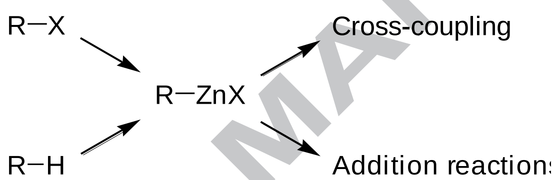

Novel methods for the preparation and reactions of organozinc reagents are reviewed.

Highlights

Generation of organozinc reagents.

Halogen-zinc exchange.

Cross-coupling of organozinc reagents.